JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 1 - Fundamentals of Environment & Sustainability

Module Overview

This handbook is your complete, beginner-to-advanced guide to the **Environment and Sustainability** — what the environment is, how it sustains life, and why every citizen has a stake in protecting it. Whether you are a student, a working professional, a parent, or a civic volunteer, this handbook is written so you can read it once and walk away with a clear, practical understanding of the fundamentals.

By the end of this module, you will not need to search anywhere else to understand the foundations of environmental science and sustainable development.

Learning Objectives

After completing this module, you will be able to:

- Define the environment and explain its biotic and abiotic components
- Explain what an ecosystem is and why biodiversity matters
- Describe the goals and three pillars of sustainable development
- Identify the major environmental challenges facing India and the world
- Understand how human activity impacts natural systems
- Recognise the government bodies and laws that protect India's environment
- Apply basic environmentally responsible habits in daily life
- Know exactly where and how to report an environmental violation in India

Why This Topic Matters

India is home to over 1.4 billion people and one of the world's richest, most varied natural landscapes — from Himalayan glaciers to coastal mangroves. This natural wealth provides the air, water, food, and raw materials that every citizen depends on daily. Yet rapid urbanisation, industrial growth, and rising consumption are placing unprecedented pressure on these systems.

> ***Reality Check:*** *Global wildlife populations have declined by an average of 73% since 1970 (WWF Living Planet Report 2024), and India's own forest cover, while slowly increasing, still falls short of the National Forest Policy target of 33% of geographical area. Environmental damage is rarely caused by one dramatic event — it is the sum of millions of small, everyday choices.*

Environmental literacy is no longer a specialist subject for scientists — it is a **life skill**, as essential as financial literacy or road safety. Understanding how ecosystems work and what sustains them is the first step toward becoming a citizen who protects rather than depletes the systems we all rely on.

This is why JanGyan begins its Environment & Civic Responsibility category with this foundational module.

Introduction

The **Environment** refers to everything that surrounds and sustains life — air, water, land, sunlight, plants, animals, and the climate system that connects them. It provides the natural resources and conditions that make human life, agriculture, and industry possible.

In simple words: the environment is the life-support system of the planet — the same way your home has walls, water supply, and electricity that keep it functioning, Earth has forests, oceans, and an atmosphere that keep all life functioning.

The environment is not a passive backdrop to human activity. It is a dynamic, interconnected system where a change in one part — the loss of a forest, the pollution of a river — ripples outward to affect climate, agriculture, and public health elsewhere.

Sustainability is the practice of using the environment's resources in a way that meets present needs without compromising the ability of future generations to meet their own needs (Brundtland Commission, 1987). Awareness of how the environment works is the first and most powerful step toward sustainability.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of environmental science is built upon.

Components of the Environment

The environment is broadly divided into two components that work together to sustain life on Earth.

Component	Description	Examples
Biotic (Living)	All living organisms that interact within the environment	Humans, animals, plants, micro-organisms
Abiotic (Non-living)	Physical and chemical factors that support life	Air, water, soil, sunlight, temperature, minerals

Key Definitions

- **Ecosystem** — A community of living organisms interacting with each other and their physical environment, functioning as a unit (a forest, a pond, a coral reef).
- **Biodiversity** — The variety of life at genetic, species, and ecosystem levels within a given area.
- **Natural Resource** — Any material or energy source (water, minerals, forests, sunlight) occurring in nature that humans use.
- **Carrying Capacity** — The maximum population an ecosystem can sustainably support without degrading.
- **Ecological Footprint** — A measure of how much land and resources an individual, city, or nation consumes relative to what nature can regenerate.

Complete Detailed Explanation

Part A — Ecosystems & Biodiversity, In Depth

An ecosystem consists of producers (plants, which convert sunlight into energy), consumers (animals that eat plants or other animals), and decomposers (fungi and micro-organisms that break down dead matter and recycle nutrients). These roles connect into food chains and food webs that keep an ecosystem in balance.

Biodiversity operates at three levels — genetic diversity within a species, species diversity within a habitat, and ecosystem diversity across a landscape. India is one of 17 “megadiverse” countries in the world and hosts four global biodiversity hotspots: the Himalayas, the Western Ghats, the Indo-Burma region, and Sundaland (Nicobar Islands).

Part B — The Ten Building Blocks of Module 1

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian citizens.

1. What is Environment?

The environment is everything — living and non-living — that surrounds us and provides the conditions for life: air to breathe, water to drink, soil to grow food, and a climate stable enough to sustain agriculture and settlement.

2. Components of the Environment

Covered above as biotic and abiotic components — together these form the physical and biological base on which every economy and society depends.

3. Ecosystems & Biodiversity

Ecosystems provide essential services — pollination, water purification, flood control, and climate regulation — that would cost enormous sums to replace artificially. Biodiversity is what keeps these services resilient to shocks like drought, disease, or pests.

4. Importance of Environmental Conservation

Conservation protects the ecological balance that supports agriculture, fisheries, clean water, and public health. It also preserves natural and cultural heritage — from wildlife sanctuaries to sacred groves — for future generations.

5. Sustainable Development

Sustainable development balances three pillars — Environmental (protecting ecosystems), Social (equity and wellbeing), and Economic (viable livelihoods) — recognising that long-term prosperity is impossible without a healthy environment.

6. Environmental Challenges

India and the world face interlinked challenges: climate change, deforestation, biodiversity loss, air and water pollution, soil degradation, and plastic waste — each intensified by population growth, urbanisation, and unsustainable consumption.

7. Human Impact on Nature

Industrialisation, deforestation, overfishing, mining, and intensive agriculture have altered ecosystems at a pace nature cannot always adapt to, leading to habitat loss, species decline, and disrupted natural cycles such as the water and carbon cycles.

8. Environmental Awareness

Awareness means understanding how everyday actions — the electricity used, the waste generated, the water consumed — connect to larger environmental outcomes. It is built through education, media, and direct community experience.

9. Global Environmental Goals (SDGs Overview)

The UN Sustainable Development Goals set measurable global targets on clean water (SDG 6), clean energy (SDG 7), sustainable cities (SDG 11), responsible consumption (SDG 12), climate action (SDG 13), life below water (SDG 14), and life on land (SDG 15) — guiding national policy through to 2030.

10. Role of Citizens in Protecting the Environment

Citizens shape environmental outcomes through everyday choices — conserving resources, reducing waste, choosing sustainable products — and through civic action: reporting violations, joining clean-up drives, and holding institutions accountable.

Important Terminologies

Term	Simple Meaning
Ecosystem	A community of living organisms interacting with their physical environment
Biodiversity	The variety of life — genetic, species, and ecosystem — in a given area
Sustainability	Meeting present needs without compromising future generations' ability to meet theirs
Carbon Footprint	The total greenhouse gases generated by a person, product, or activity
Carrying Capacity	The maximum population an ecosystem can sustainably support
Biodegradable	Material that can be broken down naturally by micro-organisms
Renewable Resource	A resource that naturally replenishes over time (sunlight, wind, water)
Non-Renewable Resource	A finite resource that does not replenish within a human lifetime (coal, oil)
Precautionary Principle	Acting to prevent environmental harm even without full scientific certainty
Afforestation	Planting trees in an area that previously had no forest cover

Step-by-Step Understanding

How an Ecosystem Sustains Life

1. **Energy Input** — Sunlight is captured by plants through photosynthesis
2. **Production** — Plants (producers) convert this energy into food and biomass
3. **Consumption** — Herbivores eat plants; carnivores eat herbivores, forming a food chain
4. **Decomposition** — Fungi and micro-organisms break down waste and dead matter
5. **Nutrient Cycling** — Nutrients return to soil, water, and air to begin the cycle again

How to Build Your Personal Environmental Responsibility

1. Understand your own resource use — water, electricity, waste generated
2. Reduce consumption before reusing or recycling
3. Choose sustainable products and services where possible
4. Support local conservation and clean-up initiatives
5. Stay informed on environmental policy and civic decisions in your area
6. Report pollution or environmental violations through official channels

Real-Life Indian Examples

(Widely known public examples, described for educational awareness.)

- **Chipko Movement (1970s, Uttarakhand):** Villagers, led largely by women, embraced trees to prevent commercial logging — a landmark example of grassroots citizen-led conservation that influenced national forest policy.
- **Silent Valley Movement (Kerala, 1970s–80s):** Sustained public and scientific opposition halted a hydroelectric project that would have submerged a rare rainforest ecosystem, leading to its declaration as a national park.
- **Namami Gange Programme:** A major national mission to clean and rejuvenate the river Ganga through pollution control, sewage treatment, and afforestation along the riverbank.
- **India State of Forest Report 2023:** India's total forest and tree cover reached 25.17% of its geographical area — a gradual increase, though still below the 33% national policy target, showing both the progress and the distance still to cover.

Practical Scenarios

Scenario 1: Your neighbourhood has started burning garbage in an open lot every evening.

This is an environmental violation and an air-pollution health hazard. Do not participate; report it to your local municipal authority or pollution control board.

Scenario 2: A local stream that once had fish now runs a strange colour and smells of chemicals.

This suggests industrial or sewage discharge. Avoid contact, note the location and time, and report it to the State Pollution Control Board.

Scenario 3: A housing project near your area plans to clear a small patch of wetland.

Wetlands regulate flooding and support biodiversity. Citizens can raise objections during the public consultation stage of the project's environmental clearance process.

Government Rules / Guidelines

- **Ministry of Environment, Forest and Climate Change (MoEFCC):** The apex national body responsible for environmental policy, forest conservation, wildlife protection, and India's climate commitments.
- **Central Pollution Control Board (CPCB):** Sets and enforces national standards for air and water quality, and coordinates with State Pollution Control Boards (SPCBs) on monitoring and compliance.
- **Forest Survey of India (FSI):** Conducts the biennial India State of Forest Report, assessing forest cover, carbon stock, and forest health nationwide.
- **National Green Tribunal (NGT):** A specialised judicial body that handles environmental disputes and violations swiftly, outside the regular court system.

Laws & Policies

Law / Policy	What It Covers
Environment (Protection) Act, 1986	Umbrella law empowering the government to regulate, inspect, and act against any activity harming the environment
Water (Prevention & Control of Pollution) Act, 1974	Regulates discharge of pollutants into water bodies and establishes pollution control boards
Air (Prevention & Control of Pollution) Act, 1981	Regulates air pollutant emissions and establishes air quality standards
Forest (Conservation) Act, 1980	Regulates diversion of forest land for non-forest use
Wildlife Protection Act, 1972	Protects wild animals, birds, and plants; regulates hunting and trade
Biological Diversity Act, 2002	Protects India's biological resources and traditional knowledge from unauthorised use
National Green Tribunal Act, 2010	Establishes the NGT for speedy resolution of environmental cases

> *The Environment (Protection) Act, 1986 was enacted in the aftermath of the Bhopal Gas Tragedy and remains the principal umbrella legislation for environmental protection in India.*

Important Statistics

Metric	Figure
India's total forest and tree cover (ISFR 2023)	25.17% of geographical area (21.76% forest + 3.41% tree cover)
National Forest Policy target for forest cover	33% of geographical area
Increase in India's forest & tree cover since 2021	1,445.81 sq. km
Global wildlife population decline since 1970 (WWF, 2024)	73% average decline
Freshwater species population decline since 1970 (WWF, 2024)	85% average decline
India's mangrove cover (ISFR 2023)	4,991.68 sq. km (about 0.15% of geographical area)
India's NDC forest carbon sink target	2.5–3.0 billion tonnes of CO■ equivalent by 2030

(Figures sourced from the India State of Forest Report 2023, Forest Survey of India / MoEFCC, and the WWF Living Planet Report 2024; official portals should be checked for the latest updates.)

Best Practices

- Reduce consumption before reusing or recycling
- Conserve water and electricity as a daily habit, not an occasional effort
- Choose products with minimal packaging and genuine sustainability credentials
- Support and participate in local afforestation and clean-up drives
- Stay informed about environmental policy decisions that affect your area
- Segregate household waste at source

Common Mistakes

- Assuming environmental protection is solely the government's responsibility
- Treating conservation as a one-time activity (a single tree-planting day) rather than an ongoing habit
- Ignoring or not reporting visible pollution or illegal dumping
- Overlooking the environmental cost of overconsumption and single-use products
- Believing that individual actions are too small to matter

Myths vs Facts

Myth	Fact
"The environment is too big for one person's actions to matter"	Collective individual habits, multiplied across millions of citizens, are what drive large-scale change
"Environmental protection slows economic growth"	Sustainable development shows long-term economic resilience depends on healthy ecosystems
"Recycling alone solves the waste problem"	Reducing and reusing prevent waste before it is created; recycling is the last, not the first, step
"Only industries pollute — households don't contribute much"	Household waste, water use, and energy consumption together form a significant share of environmental impact

Do's and Don'ts

Do	Don't
Segregate waste into wet and dry categories	Don't burn garbage or dry leaves in the open
Report pollution or illegal dumping to authorities	Don't assume someone else will report it
Conserve water and electricity daily	Don't leave taps running or lights on unnecessarily
Support local conservation and afforestation efforts	Don't treat conservation as a one-day activity
Stay informed on local environmental decisions	Don't ignore public consultation notices for local projects

Practical Tips

For Households

Segregate wet and dry waste before disposal • Fix leaking taps promptly • Switch to energy-efficient appliances where possible

For Communities

Organise or join local clean-up and plantation drives • Set up community composting where feasible • Encourage neighbours to reduce single-use plastic

For Schools & Families

Involve children in nature walks and tree-planting activities • Discuss the source of daily food, water, and energy at home • Model responsible consumption habits

Practical Checklist

■	I can explain what biotic and abiotic components of the environment are
■	I understand the three pillars of sustainable development
■	I can name at least three major environmental challenges facing India
■	I know which government body to contact to report pollution
■	I have identified one daily habit I can change to reduce my environmental footprint
■	I know where to find India's official forest and environmental data
■	I have discussed environmental responsibility with family or peers

Quick Revision Notes

- Environment = everything, living and non-living, that surrounds and sustains life
- Biotic components = living organisms; Abiotic components = air, water, soil, sunlight
- Sustainable development rests on three pillars: Environmental, Social, Economic
- India is a megadiverse country with four global biodiversity hotspots
- MoEFCC and CPCB are India's key environmental regulatory bodies
- Key laws: Environment (Protection) Act 1986, Water Act 1974, Air Act 1981, Wildlife Protection Act 1972

Key Takeaways

1. The environment is an interconnected system, not a passive backdrop to human activity

2. Sustainability means meeting present needs without compromising future generations

3. Most environmental damage results from the accumulation of everyday choices, not single events

4. India has a strong legal and institutional framework — but enforcement depends on citizen participation and reporting

5. Small, consistent habits — multiplied across a community — create meaningful environmental impact

Glossary

- **Abiotic** — Non-living components of the environment (air, water, soil, sunlight)
- **Biotic** — Living components of the environment (plants, animals, micro-organisms)
- **Biodiversity** — Variety of life at genetic, species, and ecosystem levels
- **Carbon Footprint** — Total greenhouse gases generated by a person, product, or activity
- **Carrying Capacity** — Maximum population an ecosystem can sustainably support
- **Ecosystem** — A community of organisms interacting with their physical environment
- **Megadiverse Country** — One of a small group of nations home to the majority of the world's species
- **Precautionary Principle** — Acting to prevent harm even without full scientific certainty
- **Renewable Resource** — A resource that naturally replenishes over time
- **Sustainable Development** — Development that meets present needs without compromising future generations

Frequently Asked Questions

1. What is the environment in simple words?
Everything around us — living and non-living — that provides the conditions and resources needed for life, including air, water, soil, and climate.

2. What is the difference between environment and ecosystem?
The environment is the total surrounding system; an ecosystem is a specific, localised community of organisms interacting with their physical surroundings, such as a forest or pond.

3. Why is biodiversity important?
Biodiversity keeps ecosystems resilient, supports food security, medicine, and livelihoods, and provides essential services like pollination and water purification.

4. What are the three pillars of sustainable development?
Environmental, Social, and Economic — real sustainability requires balancing all three together.

5. Is India rich in biodiversity?
Yes. India is one of 17 megadiverse countries and hosts four global biodiversity hotspots: the Himalayas, Western Ghats, Indo-Burma region, and Sundaland.

6. What is India's current forest cover?
According to the India State of Forest Report 2023, India's total forest and tree cover is 25.17% of its geographical area, still below the National Forest Policy target of 33%.

7. Which government body regulates pollution in India?
The Central Pollution Control Board (CPCB) at the national level, working with State Pollution Control Boards (SPCBs).

8. What law is India's principal environmental legislation?
The Environment (Protection) Act, 1986, enacted after the Bhopal Gas Tragedy.

9. What is the National Green Tribunal?
A specialised judicial body established under the NGT Act, 2010, for the speedy resolution of environmental disputes and violations.

10. How can an individual citizen make a real difference?
By reducing consumption, conserving resources, supporting local conservation efforts, and reporting environmental violations — habits that create meaningful impact when adopted widely.

Official References

- Ministry of Environment, Forest and Climate Change (MoEFCC) — policy and programme information
- Forest Survey of India (FSI) — India State of Forest Report 2023
- Central Pollution Control Board (CPCB) — pollution monitoring and standards
- WWF Living Planet Report 2024 — global biodiversity trends
- Environment (Protection) Act, 1986, and related environmental legislation, India Code portal
- United Nations Sustainable Development Goals — official SDG framework

Official Government Website Links

- MoEFCC: **https://moefcc.gov.in**
- Central Pollution Control Board: **https://cpcb.nic.in**
- Forest Survey of India: **https://fsi.nic.in**
- National Green Tribunal: **https://greentribunal.gov.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

Understanding the environment is not about memorising facts — it is about seeing the world as an interconnected system that sustains every part of our lives. The choices we make daily, multiplied across millions of citizens, are what ultimately determine whether India's forests, rivers, and biodiversity thrive or decline.

This is only the beginning of your Environment & Civic Responsibility journey with JanGyan. Stay informed, stay involved, and remember: sustainability begins with awareness.

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Module 1 of the Environment & Civic Responsibility Category